

INDIAN STATISTICAL INSTITUTE

Midsemestral Examination

MMath 2nd year

Algebraic Number Theory

Date: February 28, 2026

Maximum marks: 20

Time: 90 minutes

Answer any two questions.

- ✓ 1. Let R be an integral domain with field of fractions K . Assume that the set of fractional ideals of R form a group under multiplication. If I is a fractional ideal, then prove that:
- (a) $I^{-1} = \{a \in K \mid aI \subseteq R\}$. [3]
 - (b) I is a finitely generated R -module. [3]
 - (c) I is a projective R -module. [4]
- ✓ 2. Let K be a number field. For a nonzero ideal $I \subseteq \mathcal{O}_K$, the norm is defined as $\mathcal{N}(I) = [\mathcal{O}_K : I]$.
- (a) Prove that the norm is multiplicative. [5]
 - (b) Prove that $\mathcal{N}(I) \in I$. [1]
 - (c) If $\mathcal{N}(I)$ is prime, show that I is prime. Is the converse true? Justify. [1+1]
 - (d) Let P be a nonzero prime ideal. We know that P contains a unique prime number p . Show that $\mathcal{N}(P) = p^m$, where $m \leq [K : \mathbb{Q}]$. [2]
- ✓ 3. Let $K = \mathbb{Q}(\sqrt{d})$, where d is a square-free integer. Prove that $\mathcal{O}_K$ is a Euclidean domain for $d = -1, -2, -3, -7, -11$. [10]
4. Let K be a number field of degree n . Prove that $\mathcal{O}_K$ is a free abelian group of rank n . [10]